Shubhendu Jena

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GitHub Sjena ShubhenduJena

Employment History

2025 – present **Research Engineer.** CEA List, Saclay

- Contributing to SSDM (Single-Shot Deep Manta), a BEV perception framework for autonomous driving across Nurec, Argoverse, nuScenes, ZOD, ONCE, and DeepPerception datasets.
- Developed multi-task training configurations for Nurec and Argoverse, jointly optimizing 3D object detection and HD-map prediction from seven surround-view cameras.
- Implemented and validated HD-map training targets including crosswalks, road edges, continuous/discontinuous lanes, intersections and road markings using LiDAR/radially masked focal and dice losses. Used mIoU and 3D Chamfer metrics for masked and unmasked evaluation of predicted HD-maps.
- Extended deployment and inference tooling for ONNX / TensorRT backends, including in-model HD-lane coordinate decoding and world-frame 3D HD lane outputs for downstream validation.
- Built dataset, experiment, and debugging infrastructure: Lance-format data loading, Hydra/config-driven train-test pipelines, SLURM jobs, GitLab CI updates, and BEV/camera visualization tools for qualitative geometry checks.

2020 – 2022 **Research Engineer.** INRIA, Rennes
Improved on graph convolution based approaches for human shape and pose estimation from monocular input, using pixel-aligned local image features. Published in FG 2021.

2019 – 2020 **Research Assistant.** DFKI, Kaiserslautern
Worked on 3D scene-flow prediction pipelines and helped in factoring code of previous TensorFlow 1.x projects into TensorFlow2.

Education

2022 – 2025 **Ph.D. in Computer Science, INRIA/Université de Rennes, France**
Thesis: *Combining Implicit and Explicit Representations for Modeling 3D Shape and Appearance*

- Researched generalizable and scene-specific methods for novel-view synthesis and 3D reconstruction from sparse images.
- Explored explicit representations (Gaussian splatting and point-based neural rendering) and implicit representations employing volumetric rendering.
- Developed a generalizable 2D Gaussian Splatting approach by fine-tuning features from the 3D foundational model *MASt3R*.
- Worked on bundle adjusting 2D gaussian splatting approach using *MASt3R* point clouds and initial camera estimates for state-of-the-art 3D reconstruction without ground truth camera poses.
- Explored vision-language and generative models, including CLIP and Stable Diffusion.

2017 – 2020 **M.Sc. in Visual Computing, Universität des Saarlandes, Germany**
Thesis: *End-to-End 3D Scene Flow Prediction from LiDAR Point Clouds and RGB Data*

- Designed an end-to-end neural network for dense 3D scene flow estimation using multimodal fusion of point clouds and RGB data.
- Introduced a novel RGB aggregation scheme for enhanced feature extraction.
- **Relevant Courses:** Image Processing and Computer Vision, Computer Graphics, Elements of Statistical Learning.

2012 – 2017 **Dual Degree in Electrical Engineering, NIT Rourkela, India**
Thesis: *Detection of Cerebral Palsy Activity from Video*

- Classified motion trajectories of infants as healthy or affected using methods like K-Nearest Neighbors, Logistic Regression, and Support Vector Machines.
- **Relevant Courses:** Digital Image Processing, Digital Signal Processing.

Research Publications

1. **S. Jena**, A. Ouasfi, M. Younes, and A. Boukhayma, "Sparfels: Fast reconstruction from sparse unposed imagery," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, Oct. 2025, pp. 27 476–27 487.
2. **S. Jena**, S. Reddy, and A. Boukhayma, "Sparsplat: Fast multi-view reconstruction with generalizable 2d gaussian splatting," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR Workshop)*, Jun. 2025.
3. A. Ouasfi, **S. Jena**, E. Marchand, and A. Boukhayma, "Toward robust neural reconstruction from sparse point sets," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2025, pp. 6552–6562.
4. **S. Jena**, F. Multon, and A. Boukhayma, "Geotransfer : Generalizable few-shot multi-view reconstruction via transfer learning," in *European Conference on Computer Vision (ECCV Workshop)*, Springer, 2024.

5 **S. Jena**, F. Multon, and A. Boukhayma, "Neural mesh-based graphics," in *European Conference on Computer Vision (ECCV Workshop)*, Springer, 2022.

6 **S. Jena**, F. Multon, and A. Boukhayma, "Monocular human shape and pose with dense mesh-borne local image features," in *2021 16th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2021)*, IEEE, 2021.

Skills

Languages	Français (B1), English (C1).
Coding	Python, C++, Bash, SQL, R, LaTeX
Libraries	PyTorch, PyTorch Lightning, TensorFlow, Hugging Face, NumPy, SciPy, Scikit-learn, OpenCV, MMCV / MMDetection
Autonomous driving	BEV perception, multi-camera 3D object detection, HD-map segmentation, Gaussian splatting / rasterization, LiDAR-camera geometry, Argoverse, nuScenes
Tools	Hydra configs, Lance datasets, ONNX, TensorRT, CUDA, Docker, SLURM, GitLab CI, Unix/Linux, Git, GitHub
Misc.	GRE - 328/340, TOEFL iBT - 110/120.

Miscellaneous Experience

Awards and Achievements

2012, 2013 **Merit Award**, Top 3 in batch, dual degree Electrical Engineering.

2025 **ICCV Highlight paper** - Sparfels: Fast reconstruction from sparse unposed imagery,.

Teaching

2019 **Tutor, Image Processing and Computer Vision**. MIA group, Saarland University.

References

Available on Request